Introduction. My research interests lie in social computing and computational social science. At the core of my ethos as a researcher is advancing inclusive AI technologies and online platforms to better serve people who are overlooked in the development of these systems. Driven by my *interdisciplinary* background in computer science and political science, I aim to develop technical methods grounded in social science theories to address social-oriented questions with two primary goals:

- **Social Computing:** Auditing AI technologies and online platforms for bias and misinformation to build a more responsible sociotechnical future.
- **Computational Social Science:** Understanding social phenomena through language and behaviors to promote socially attuned and equitable technologies.

At the University of Washington, I have been fortunate to pursue these goals under the guidance of Professors Tanu Mitra, Yulia Tsvetkov, and David Jurgens, and to author four papers at EMNLP and ICWSM.

Social Computing. I aim to address social inequities in large-scale online systems like YouTube [1, 2] and LLMs [3, 4, 5]. One issue I tackled was COVID-19 misinformation. While YouTube serves as a key health information source, it has faced criticism for spreading misinformation, undermining public health efforts globally.

Despite the global aspect of the pandemic, most algorithmic audit studies focus on the Global North context, overlooking the Global South. To address this, in my first-author paper at *ICWSM 2025* [1], I conducted a large-scale geolocation-based audit of YouTube, comparing the prevalence of COVID-19 misinformation in search results between the United States (US) and South Africa (SA). I programmed sock-puppet bots (emulating "real" users) to collect 915K YouTube search results and developed annotation heuristics based on Google's misinformation policy to label videos. Using the labeled data, I built 62 classifiers to detect COVID-19 misinformation in videos based on their metadata (e.g., titles, transcripts). Since simpler models like SVMs underperformed, I gathered additional training data and experimented with advanced models like DeBerta-base-v3 and GPT-4, achieving 85% test accuracy with the best model. Our analysis revealed that nearly 32% of the top-10 search results contained COVID-19 misinformation, with sock-puppets in SA exposed to *significantly more misinformative search results* than their US counterparts. This study revealed stark regional disparities in YouTube's algorithmic behaviors and the need for equitable content moderation globally. Currently, my ongoing research measures the prevalence of opioid use disorder misinformation on YouTube (to be submitted at *FAccT 2025* [2]) and in LLM (accepted at *ICWSM 2025* [3]).

Potential Future Work: One direction I am excited about is conducting a *multilingual* audit of video search engines to measure the prevalence of misinformation in search results across search queries in *multiple languages*. Although most content on YouTube is in non-English languages, algorithmic audits primarily focus on English. This creates biased algorithmic governance that overlooks misinformation risks in other languages, disproportionately impacting non-English-speaking communities. Conducting multilingual audits is essential to provide actionable insights for platform accountability in diverse linguistic contexts and ensure broader, more *inclusive* algorithmic governance that considers all languages.

Computational Social Science. Despite advancements in LLMs, social science theories remain underutilized in NLP. These theories offer a rich foundation for understanding human behaviors and social contexts. Integrating these theories into NLP can lead to more context-aware, robust, and socially attuned models. I am interested in adapting social science theories with language technologies to enhance our understanding of social phenomena and people through language and online interactions.

In my co-first author paper published at *EMNLP Findings 2024* [6], I co-led a project investigating how LLMs can measure implicit norms and values within online communities. Prior works focused on explicit rules (i.e., *active norms*), but most social norms are *implicitly* revealed through social interactions. Using the Return Potential Model theory from sociology, I proposed using Reddit's *upvote* system to capture *community preferences* across behaviors like sarcasm and humor. For example, in *r/wallstreetbets* (e.g., community for sarcastic discussion of stocks), sarcastic comments are often upvoted, while serious ones are downvoted, revealing the community's implicit norms. Using these theoretical insights, we developed *ValueScope*, a technical framework leveraging LLMs to model community preferences across linguistic and stylistic variations in online communities. My contributions include applying linguistic style transfer for controlled simulations of community language across norms (e.g., genuine to sarcastic), developing a data filtering pipeline, and conducting human and LLM-as-a-judge evaluations to ensure data quality. Our work showed that even closely related communities, such as *r/wallstreetbets* and *r/wallstreetbetsnew*, maintain distinct norms.

Potential Future Work: Theory-driven frameworks like *ValueScope* can inform the development of socially attuned and equitable technologies. One potential direction that I am excited about is building content moderation systems that align with a community's implicit norms. By leveraging the implicit norms identified by *ValueScope* [6], we can fine-tune models to evaluate whether a comment aligns with the implicit norms and warn users about posting potentially misaligned content. Such model can encourage more pro-social behavior that reflects the community's norms and values. In particular, the model can support moderators in fostering empathetic, stigma-free environments in online mental health and substance use recovery communities where individuals seek support.

Why Princeton. Princeton's commitment to interdisciplinary research for societal impact aligns strongly with my research interests. I am drawn to **Professor Manoel Horta Ribeiro**'s work in computational social science and social computing, which resonates with my interest in improving algorithms on online platforms and using LLMs to understand human behaviors. His works has informed my research [1, 2] since my undergraduate years. I would also love to work with **Professor Aleksandra Korolova**. Her research toward understanding the societal impacts of algorithms and AI through audits aligns with my research interests, including my prior works of online platforms and LLMs [1-5]. I am also interested in **Professor Danqi Chen**'s research on improving the factuality and trustworthiness of LLMs, particularly her paper on enabling LLMs to generate texts with citations, which has shaped my ongoing research [4]. Overall, Princeton offers the ideal environment to pursue a PhD, advance my research goals, and prepare for my career aspiration as a professor.

References

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[6] Chan Young Park*, Shuyue Stella Li*, **Hayoung Jung***, Svitlana Volkova, Tanu Mitra, David Jurgens, and Yulia Tsvetkov. ValueScope: Unveiling Implicit Norms and Values via Return Potential Model of Social Interactions. In *Proc. of EMNLP Findings 2024*. ***(* Equal Contribution)***